Hemasri Sai Lella

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Education

Indian Institute of Science, Bangalore — August 2024 – Present
MTech (res) in Computer Science and Automation — *8.7/10.0 CGPA*

Indian Institute of Technology Tirupati — November 2020 – May 2024
Bachelor of Technology in Computer Science and Engineering — *8.42/10.0 CGPA*

Technical Skills

Programming: Python, C/C++, Java, SQL
Machine Learning: Supervised Learning, Deep Learning, NLP, Transformers, RAG
Libraries and Frameworks: NumPy, Pandas, scikit-learn, PyTorch, XGBoost, FAISS, Matplotlib
Mathematical Foundations: Linear Algebra, Probability, Statistics, Optimization, Risk Minimization
Tools and Platforms: Git, Jupyter, Databricks, Oracle SQL, Microsoft SQL Server, PostgreSQL, MongoDB

Relevant Coursework

Mathematical Foundations of Machine Learning, Natural Language Processing, Generative AI: Principles and Applications, Modeling and Simulation, Program Analysis and Verification, Database Management Systems

Research and Professional Experience

Student Researcher — January 2025 – Present
Jayant R. Haritsa, DSL Lab — *Bengaluru, India*
- Building an ML-based SQL query optimization system that uses an XGBoost classifier to predict whether complex pattern-matching queries should be rewritten for better performance.
- Contributing to a safe rewrite workflow that generates, validates, and selects optimized query alternatives while preserving query correctness.

Undergraduate Student Researcher — July 2023 – May 2024
Dr. Sridhar Chimalakonda, Risha Lab — *Tirupati, India*
- Designed and developed DBJoules, a tool to measure the energy consumption of querying database systems.
- Conducted empirical evaluations across MySQL, PostgreSQL, MongoDB, and Couchbase to identify efficient database systems for different query types.
- Resulted in a peer-reviewed publication at EASE 2024.

Software Intern — May 2023 – July 2023
MAQ Software — *Hyderabad, India*
- Developed a natural language chatbot for querying a database using C# and React.
- Conducted usage growth analysis and developed Power BI dashboards to support data visualization and business insights.

Research Intern — May 2022 – July 2022
Indian Institute of Technology Madras
- Analyzed graph algorithm runtimes by comparing CPU and GPU performance using CUDA and C++.

Selected ML Projects

GPT-2 from Scratch — Ongoing
- Implementing a GPT-2 style Transformer language model from scratch with token embeddings, positional embeddings, multi-head self-attention, feed-forward blocks, and autoregressive text generation.

LLM Robustness Evaluation: Sycophancy and Prompt Sensitivity — May 2026
- Built reproducible evaluation pipelines to study behavioral reliability failures in open instruction-tuned LLMs under user-pressure and prompt-style variations.
- Evaluated Qwen2.5-7B-Instruct and Llama-3.1-8B-Instruct on 1,480 responses using accuracy, corrected flip rate, answer consistency, fragility, and explanation-similarity metrics.

Image Classification using MLP, CNN, and Vision Transformer from Scratch — May 2026
- Designed and implemented MLP and CNN architectures from scratch in NumPy, including modular vectorized backpropagation, for CIFAR-10 image classification.

English-to-Hindi Neural Machine Translation using Seq2Seq Models — May 2026

- Implemented an attention-based LSTM encoder-decoder model with Byte-Pair Encoding for English-to-Hindi name transliteration.

ASHA Sahayak April 2026

Links: GitHub | Devpost

- Built a Databricks-native maternal healthcare assistant for ASHA workers to track pregnancies, ANC visits, pathology tests, patient history, and risk alerts.
- Developed multilingual AI and patient-aware RAG components using Indic-language voice/translation support, FAISS-based retrieval, and maternal health guidelines.

Stock Market Prediction August 2024 – December 2024

- Built a continuous HMM-based stock forecasting model using OHLC features, AIC/BIC model selection, and rolling re-training; achieved ~1.1%–2.2% MAPE on Yahoo Finance data.

Interval & Pointer Analysis Tool August 2024 – December 2024

- Implemented intra-procedural interval analysis for Java bytecode using Kildall's fixed-point algorithm.
- Built a flow-sensitive points-to analysis for integer arrays and integrated it with interval analysis to detect array index-out-of-bounds errors.

Honors and Awards

- **2nd Place, Bharat Bricks Hacks 2026** – Won for ASHA Sahayak, a Databricks-deployed multilingual maternal healthcare assistant. (April 2026)
- **NPTEL Mathematical Foundations of Machine Learning** – Elite Silver Certification; Top 1%; Score: 87%. (January 2026 – May 2026)
- **Reliance Foundation Postgraduate Scholarship** – Selected among 100 postgraduate scholars across India. (March 2025)

Publications

Lella, H.S., Chattaraj, R., Chimalakonda, S. and Kurra, M., 2024, June. Towards Comprehending Energy Consumption of Database Management Systems-A Tool and Empirical Study. In Proceedings of the 28th International Conference on Evaluation and Assessment in Software Engineering (pp. 272–281).

Position of Responsibility

1. Lead Volunteer for ACM SIGMOD 2026 held in Bangalore.
2. Teaching Assistant for E0 240: Modeling and Simulation. (August 2025 – December 2025)
3. Department Representative and Student Representative for the CSA department at IISc.

Hackathons and Workshops

1. Attended Maitreyee: IBM Research India Annual Student Outreach Event. (September 2025)
2. Attended Present and Future Computing Systems Workshop at IISc. (December 2024)
3. Attended Adobe-IISc Research Workshop. (November 2024)
4. Participated in the Mercuria Hackathon. (November 2024)
5. Attended Theoretical Computer Science Winter School at IIT Delhi. (December 2022)